

- 13** Catechol is a chemical found in a number of fruits. Catechol can be oxidised to a quinone by the enzyme catechol oxidase.

Catechol oxidase is inhibited by parahydroxybenzoic acid (PHBA), which is structurally similar to catechol. In the presence of PHBA, the Michaelis-Menten constant (K_m) for catechol oxidase increases.

How does PHBA inhibit catechol oxidase?

- 1 PHBA and catechol can both bind to the enzyme but not at the same time.
- 2 PHBA is a non-competitive inhibitor of catechol oxidase.
- 3 PHBA decreases the V_{max} of the reaction.

A 1, 2 and 3 **B** 1 only **C** 2 only **D** 3 only